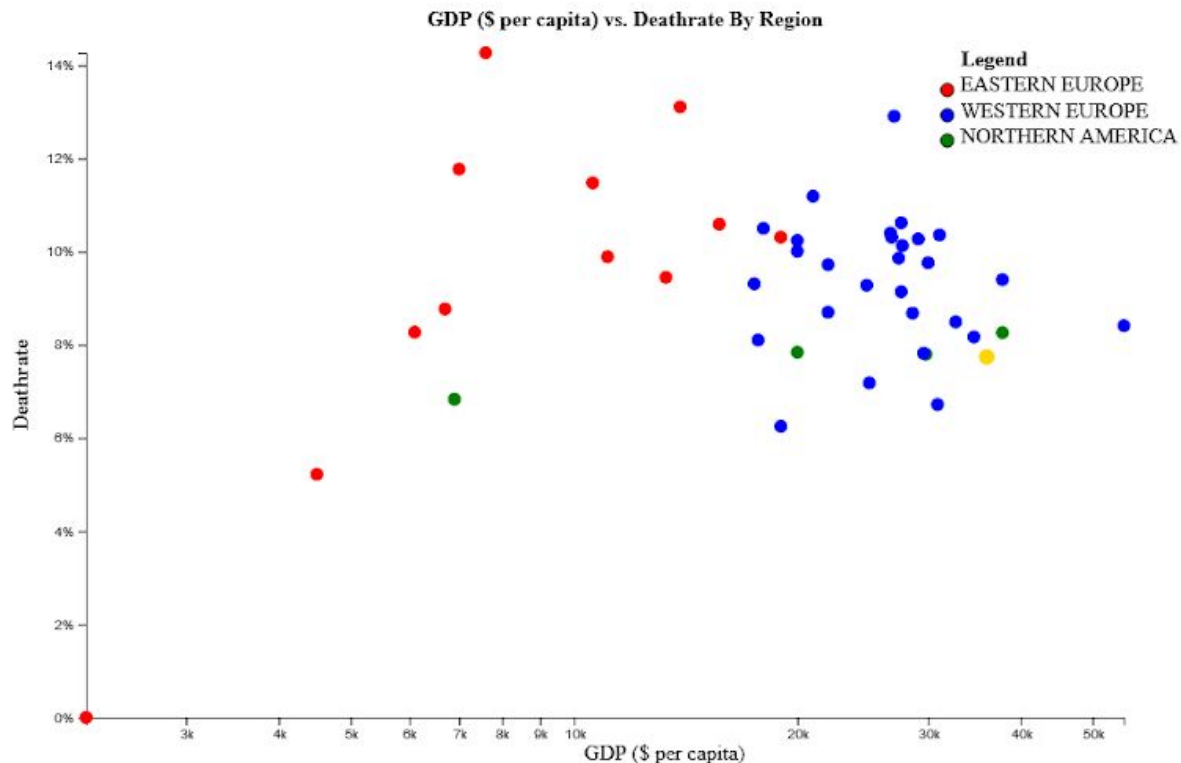


Homework #5

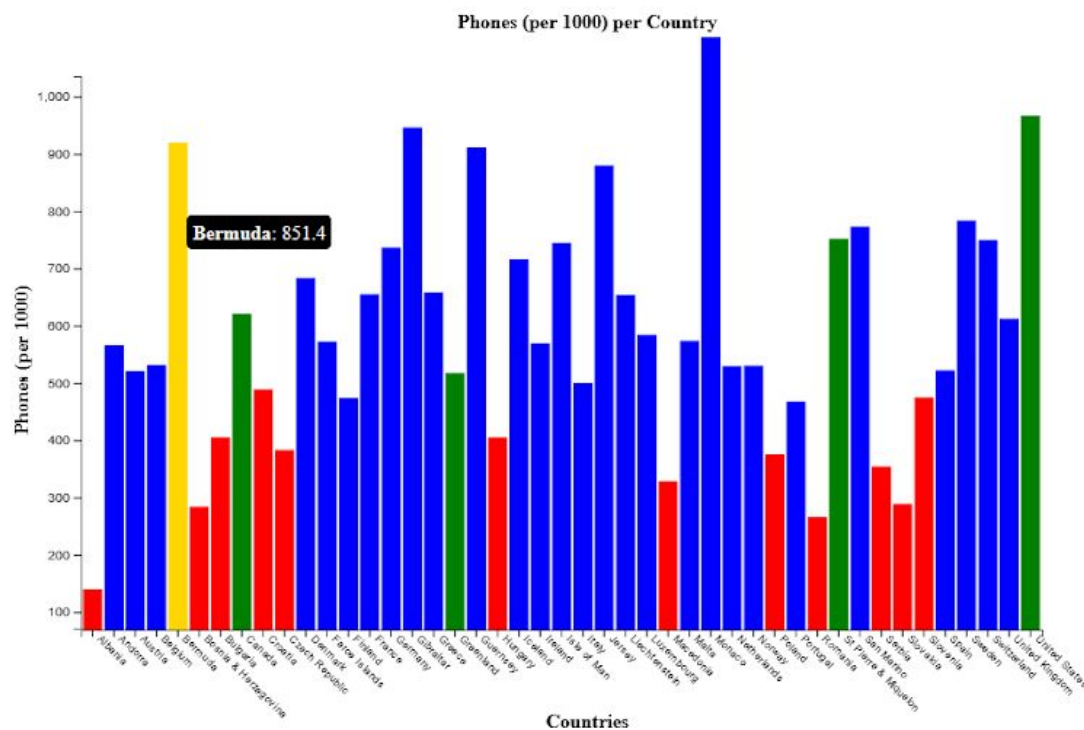
By: Gabriel Vazquez

Visualizations

To visualize the "Countries of the World" dataset, I examined the dataset attributes to see what seems interesting to visualize. I decided that I wanted to see how a country's GDP and their death rate are related. In addition, since the dataset was vastly large, I decided to restrict my analysis to countries located in Eastern Europe, Western Europe, and Northern America. For these three regions, I also wanted to see if the availability of phones had any effect on a country's death rate as a result. To visualize these attributes, I decided upon using a scatterplot to show the relationship between death rate and GDP and I used a bar graph to visualize the amount of phones in each country per a thousand people. By using a scatterplot, the user can easily identify patterns among our regions, but also see any trends *within* each of these regions. For example, Eastern Europe might have a large death rate as a region, but by closely looking at each point on the graph, the user can see how each region's countries behaves.



By using a bar graph to visualize the availability of phones in each country, the user can quickly analyze which countries have more or less phones available per 1000 people. Alone, this visualization does not provide any interesting information aside from phone availability, but combined with our scatterplot, we can probe our dataset further. For example, we can use the scatterplot to try and understand why countries who have high GDP might have a high death rate. By referring back to our bar graph, we can see if those countries have a high or low availability of phones in their country. If they have a low availability, then perhaps the citizens of that country might not have access to emergency services. We might ask something similar if they have a high availability.



Marks and Channels

Scatterplot

To encode our data into our scatterplot, I used two categorical data attributes, region and country. My marks for the scatterplot are points and my channels for these points are shape, position, color, and size. Each mark on the scatterplot represents a country that is currently in our dataset. These marks are in the shape of a circle. The color of each mark represents which region that country belongs to. If the user's mouse is hovering over a specific point, the color of the mark

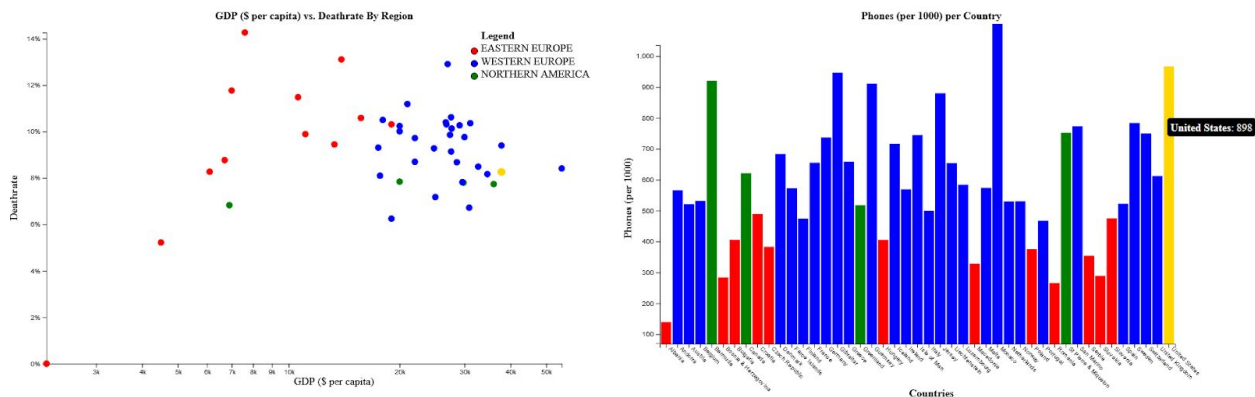
will change to gold to signify which country is selected. In addition, the country that is selected and golden will have their size slightly enlarged to help users see what they are currently selecting.

Bar Graph

The marks on my bar graph are also points, however my channels are shape, size and color. Similar to our scatterplot, each mark represents a country in our dataset, but the shape of each point is a bar (rectangle). The color is also used to identify which region a country belongs to and if it is selected or not.

Interactions

To make my visualizations interactive, I decided to include a tooltip for users to use when trying to examine specific data points. When the tooltip appears, the country's name is listed along with its corresponding GDP and death rate value. When they are examining the data point, the point changes color and size to signify that point is in focus. At the same time, the same country is colored golden on the bar graph to help users identify the country's phone availability quickly. If the users want to see the specific number of phones per country (per 1000 people), a tooltip will appear to provide them with that information. With these interactions, whether the user is interacting with the scatterplot or bar graph, they can easily identify data regarding the other chart's information for a specific country. For example, if I see that the United States have a high amount of phones per 1000 people, I can highlight the bar, and I can easily check the scatterplot for the United States' information.



Color Justification

Since there are many countries that belong to vastly different regions, if I were to not use any color to categorize each country, then it would be extremely difficult to analyze the data for patterns. By using distinct colors throughout both visuals, then it is much easier to identify patterns about regions or specific countries.